Advanced Computer Architecture
CS-2..
Lecture week 4~5: Memory Architecture and Access

Dr. Muhammad asif
Bahauddin Zakariya University
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Last Week Review

The Pentium Processor

- Pentium registers
  - Pentium has ten 32-bit registers and six 16-bit registers
    - Internal to the processor & Provide faster access
  - These registers are grouped into
    - General registers
      - Data registers e.g., AX, BX, CX, DX
      - Pointer registers e.g., BP, SP, IP
      - Index registers e.g., SI, DI
    - Control registers e.g., Flags, CPU registers
    - Segment registers e.g., CS, DS, SS, ES

ISA

- Instruction Groups
  - Data Movement Instruction
  - Arithmetic and Logical Instructions
  - Special Instructions: to change the processor state or behavior
- Instruction Set Classes e.g., 4/3/2/1/0-address instructions
- Physical Address Calculation
- Addressing Modes (Where are operands located?)
  - Register
  - Immediate
  - Direct
  - Indirect
  - Displacement
- Computer Taxonomy

Lecture 02

The Pentium Processor

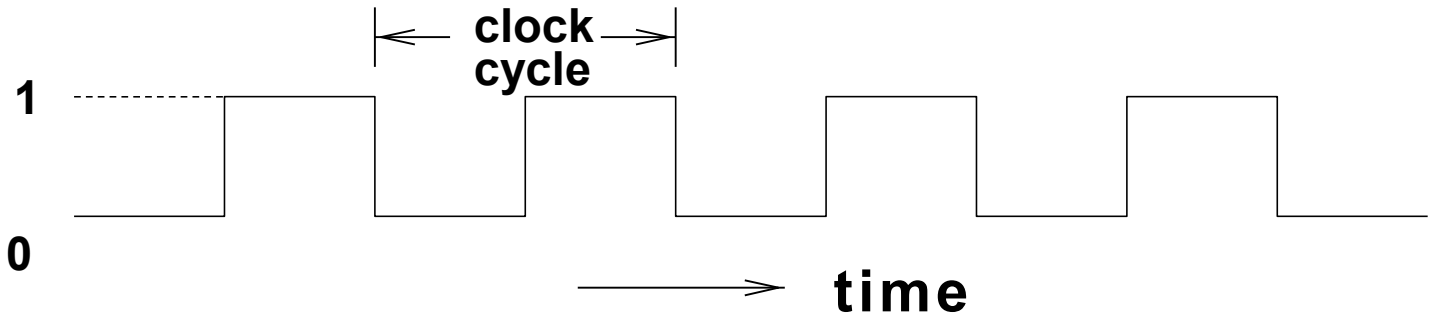
- System clock

» System clock provides timing signal to synchronize the operations of the whole system

∗ Clock period = length of time taken by one clock cycle

clock period = 1/clock rate

∗ Example: A 200 MHz clock yields a clock period of $1/(200 * 10^6) = 5$ ns



Intel 80X86 Family Processors

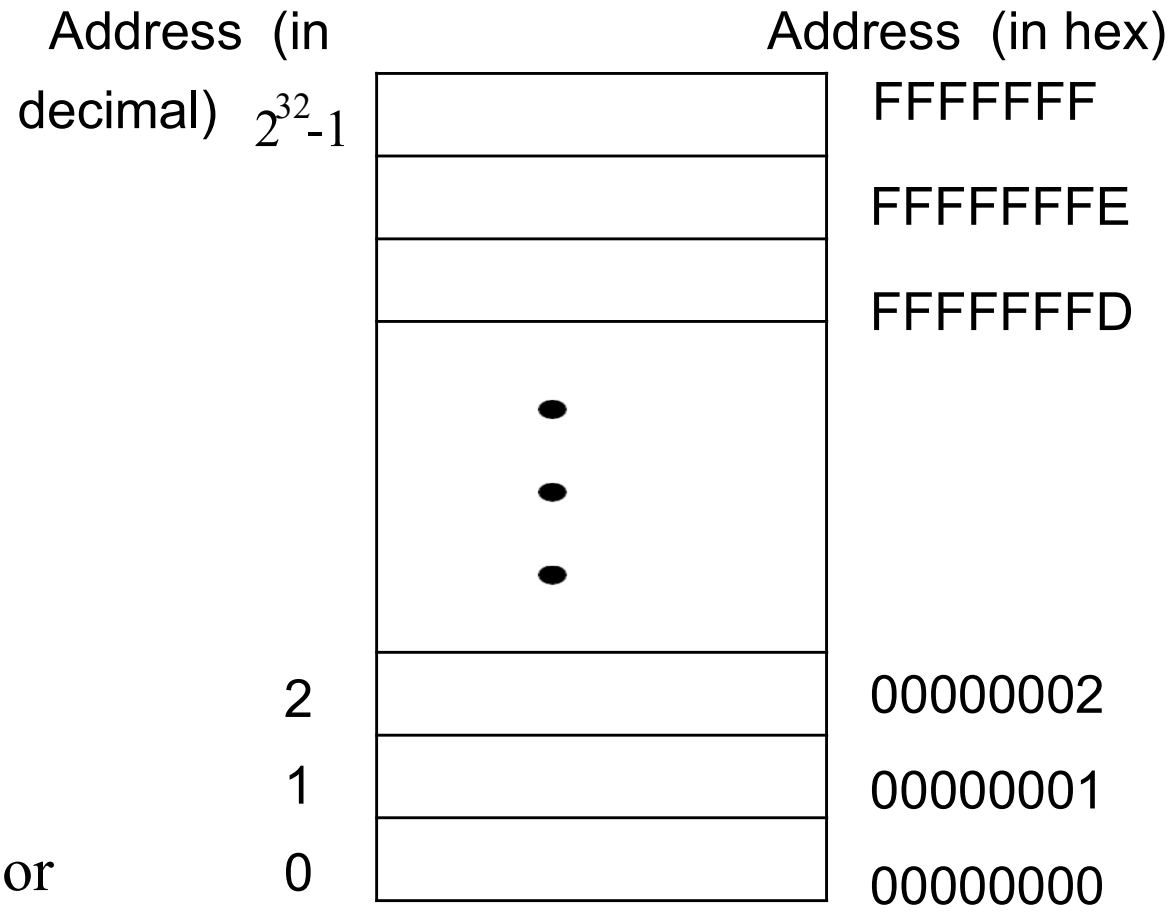
- 8086 in 1979
  - 20-bit address bus (1MB)
  - 16-bit data bus
  - No floating-point coprocessor
- 80286 in 1982
  - 24-bit address bus (16MB)
  - 16-bit data bus
  - No floating-point coprocessor
  - Memory protection capabilities
- 80386 in 1985
  - First 32-bit processor
  - 32-bit address bus (4GB)
  - 32-bit data bus
  - No floating-point coprocessor
- 80486 in 1989
  - Improved version of 386
  - 32-bit address bus (4GB)
  - 32-bit data bus
  - Includes floating-point coprocessor and internal cache

Intel 80X86 Family Processors (cont'd)

- Pentium in 1993
  - Enhanced version of 486
  - 32-bit address bus (4GB)
  - 64-bit data bus
  - All internal registers are 32-bit wide
    - Still a 32-bit processor
    - All instructions operate on at most 32-bit operands
  - Several versions
    - Some versions are stripped down for under $1000 PC market
    - Some enhanced with MMX technology

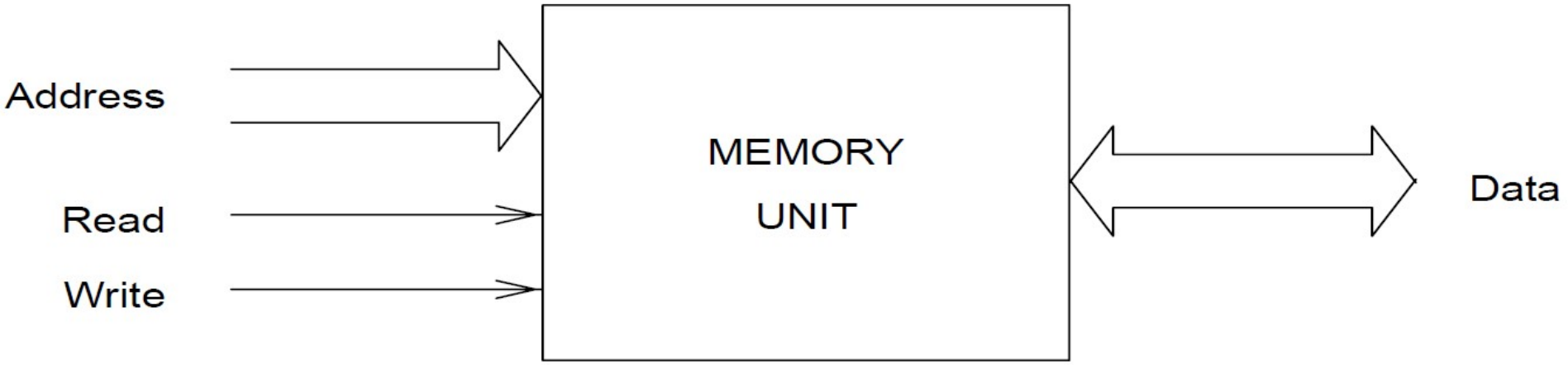
Memory

- Memory can be viewed as an ordered sequence of bytes
- Each byte of memory has an address
  - Memory address is essentially the sequence number of the byte
  - Such memories are called *byte addressable*
  - Number of address lines determine the memory address space of a processor



Memory (cont'd)

- Two basic memory operations
  - » Read operation (read from memory)
  - » Write operation (write into memory)
- Access time
  - » Time needed to retrieve data at addressed location
- Cycle time
  - » Minimum time between successive operations



Memory (cont'd)

- Steps in a typical read cycle

1. Place the address of the location to be read on the address bus
2. Activate the memory read control signal on the control bus
3. Wait for the memory to retrieve the data from the addressed memory location
4. Read the data from the data bus
5. Drop the memory read control signal to terminate the read cycle

* A simple Pentium memory read cycle takes 3 clocks

– Steps 1&2 and 4&5 are done in one clock cycle each

* For slower memories, *wait cycles* will have to be inserted

Memory (cont'd)

- Steps in a typical write cycle

1. Place the address of the location to be written on the address bus
2. Place the data to be written on the data bus
3. Activate the memory write control signal on the control bus
4. Wait for the memory to store the data at the addressed location
5. Drop the memory write control signal to terminate the write cycle

* A simple Pentium memory write cycle takes 3 clocks

– Steps 1&3 and 4&5 are done in one clock cycle each

* For slower memories, *wait cycles* will have to be inserted

Memory (cont'd)

- Some properties of memory
  - Random access
    - Accessing any memory location takes the same amount of time
  - Volatile memory
    - Needs power to retain the contents
      - Non-volatile memory
    - Retains contents even in the absence of power
- Basic types of memory
  - Read-only memory (ROM)
  - Read/write memory (RAM)

Memory (cont'd)

- Read-only memory (ROM)
  - » Cannot be written into this type of memory
  - » Non-volatile memory
  - » Most are factory programmed (i.e., written)
  - * Programmable ROMs (PROMs)
    - » Can be written once by user
      - – A fuse is associated with each bit cell
      - – Special equipment is needed to write (to blow fuse)
    - » PROMs are useful
      - – During prototype development
      - – If the required quantity is small

Memory (cont'd)

* Erasable PROMs (EPROMs)
» Can be written several times
» Offers further flexibility during system prototyping
» Can be erased by exposing to ultraviolet light
  – Cannot erase contents of selected locations
* Electrically erasable PROMs (EEPROMs)
» Contents are electrically erased
» No need to erase all contents
  – Typically a subset of the locations are erased as a group
  – Most EEPROMs do not provide the capability to individually erase contents of a single location

Memory (cont'd)

- Read/write memory

» Commonly referred to as random access memory (RAM)

» Volatile memories

* Two basic types

» Static RAM (SRAM)

– Retains data with no further maintenance

– Typically used for CPU registers and cache memory

» Dynamic RAM (DRAM)

– A tiny capacitor is used to store a bit

– Due to leakage of charge, DRAMs must be *refreshed* to retain contents

Memory (cont'd)

* SRAM versus DRAM
  - SRAMs are expensive but faster
    - Typical access time in 10-20 ns range
    - Cost in $200-400 per MB
  - DRAMs are cheaper but slower
    - Typical access time in 50-100 ns range
    - Cost in $30-100 per MB
    - Typically used as main memory
      - Due to the cost advantage

Pentium Memory Architecture

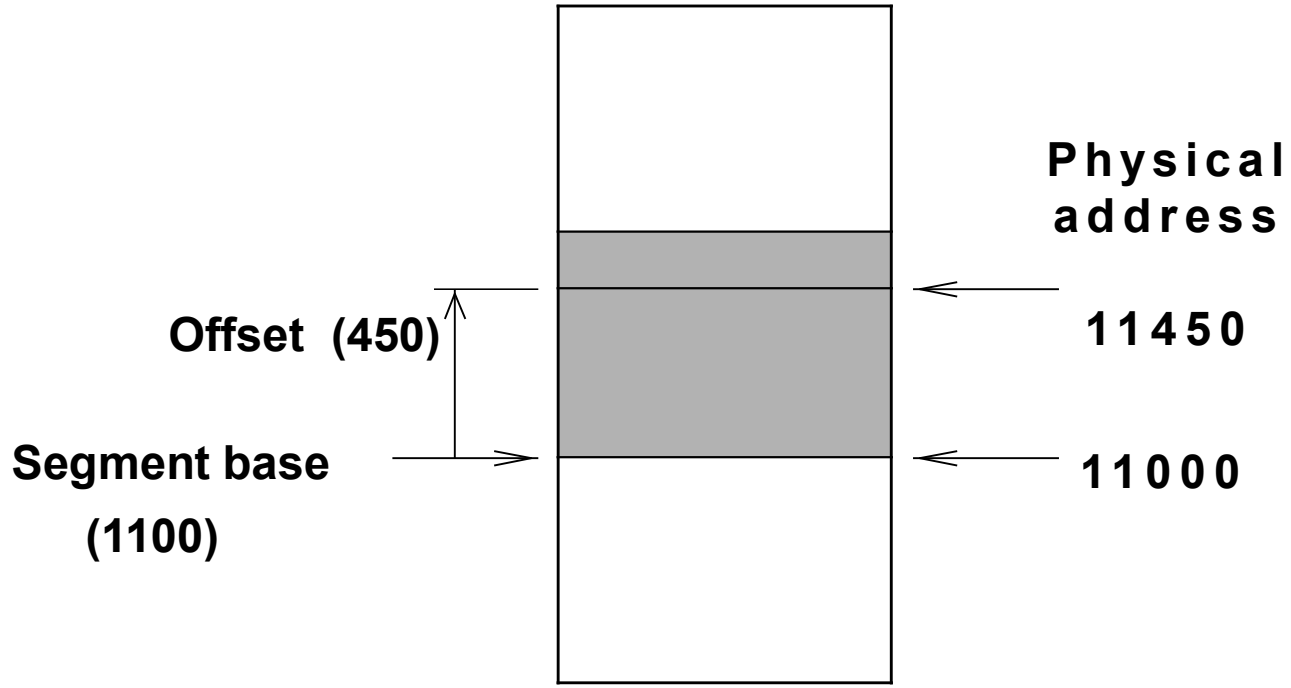
- Two modes
* Real mode
  - » Uses 16-bit addresses
  - » Supports segmented memory architecture
  - » Provides backward compatibility
    - – To run 8086 programs
* Protected mode
  - » Native mode of Pentium
  - » Uses 32-bit addresses
  - » Supports segmentation and paging

Real Mode Memory Architecture

- Real mode memory architecture
  - Pentium operates like a faster 8086 processor
  - The 8086 has
    - » 20 address lines (can address 1 MB)
    - » All registers are 16-bit wide
  - Memory is organized as segments of (up to) 64 KB
    - – Due to 16-bit registers (2^{16} = 64 K)
  - Two components are required to specify a location
    - » Segment base address (segment start address)
    - » Offset within a segment
    - » Both are 16-bit numbers

Real Mode Memory Architecture (cont'd)

* 16-bit segment base address restricts segments to start at addresses that are multiple of 16
  – Segments can only start at addresses 0, 16, 32, 48, ...



Real Mode Memory Architecture (cont'd)

- Logical address consists of two 16-bit components

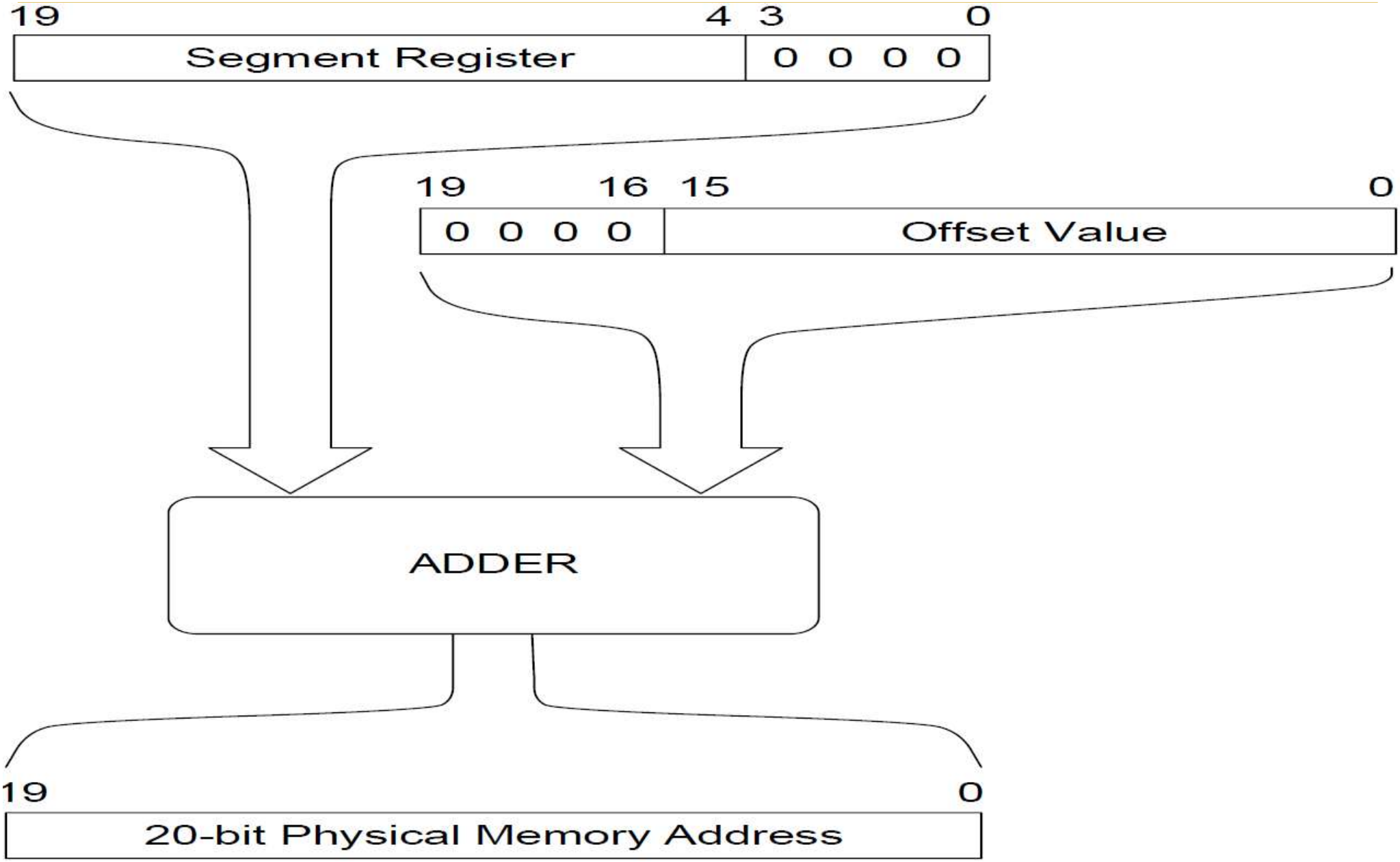
`segment:offset`

- Physical address is a 20-bit number
- Translation from logical to physical address:
  - » Append four least significant zeros to the segment base address
    - – Just add a zero when using hex numbers
      - » Add the offset value
- Example: Translate logical address 1100:450

`11000` (add zero to segment address)
`+ 450` (offset value)

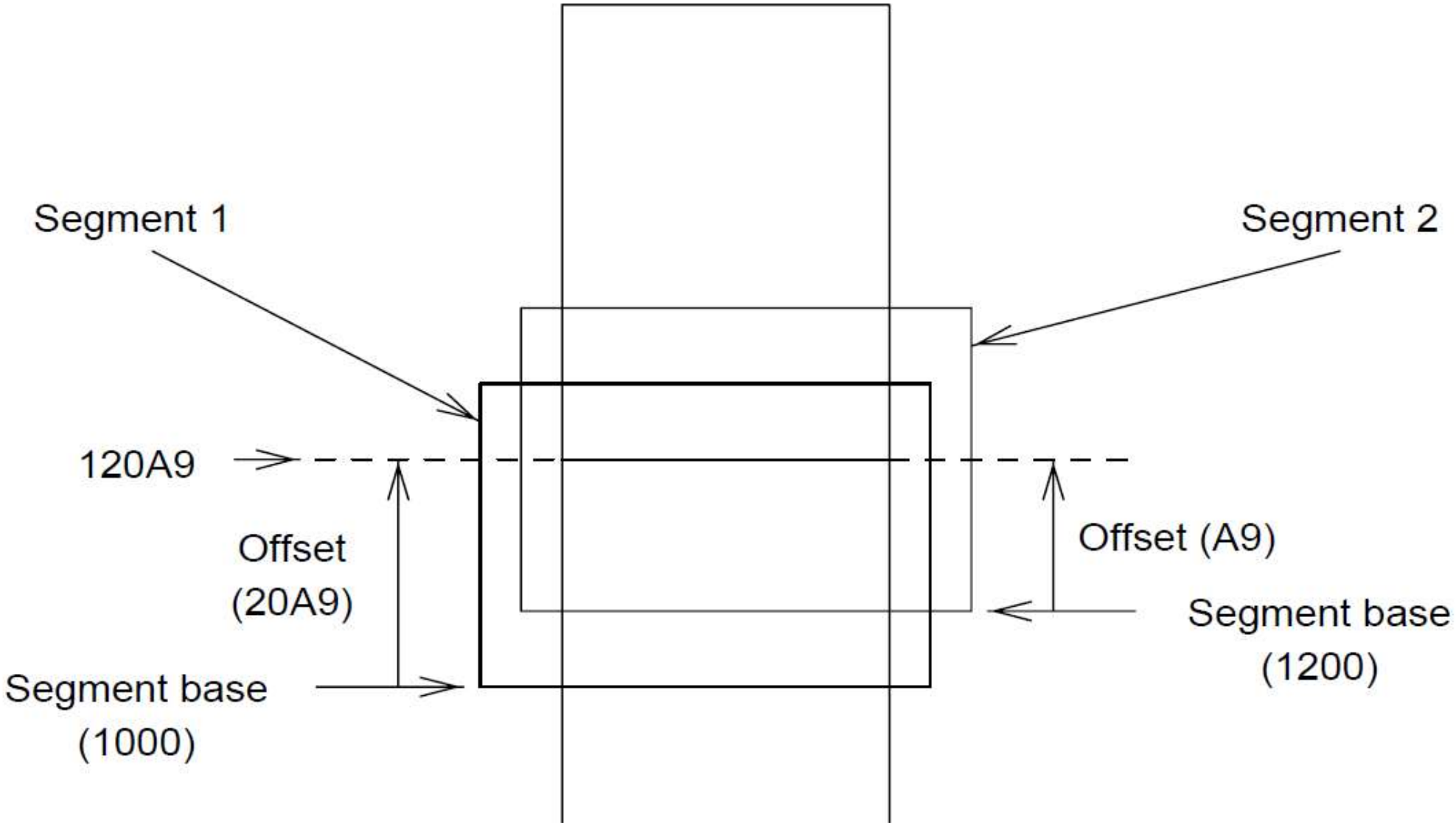
`11450` (20-bit physical address)

Real Mode Memory



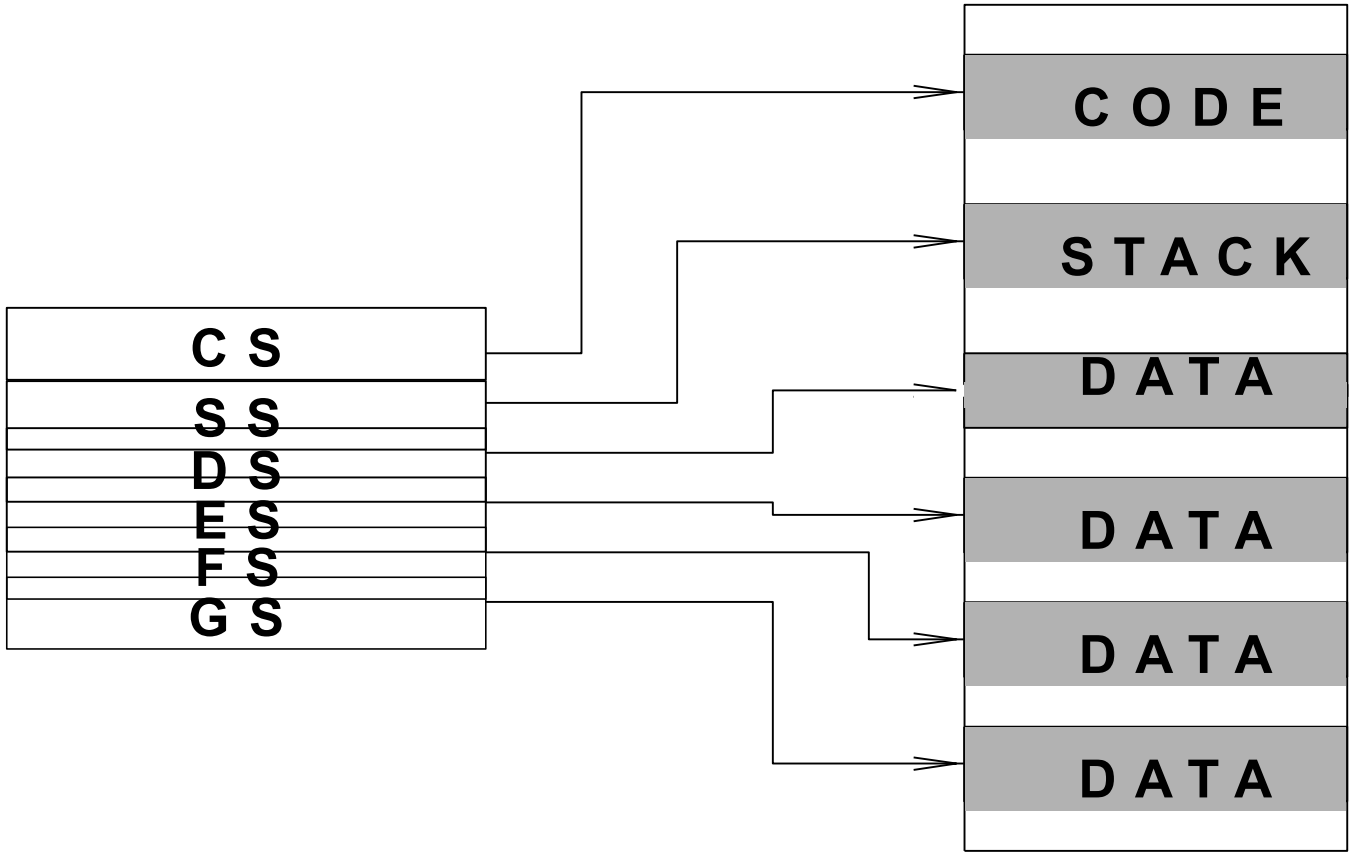
Real Mode Memory Architecture (cont'd)

Multiple logical addresses can map to the same physical address



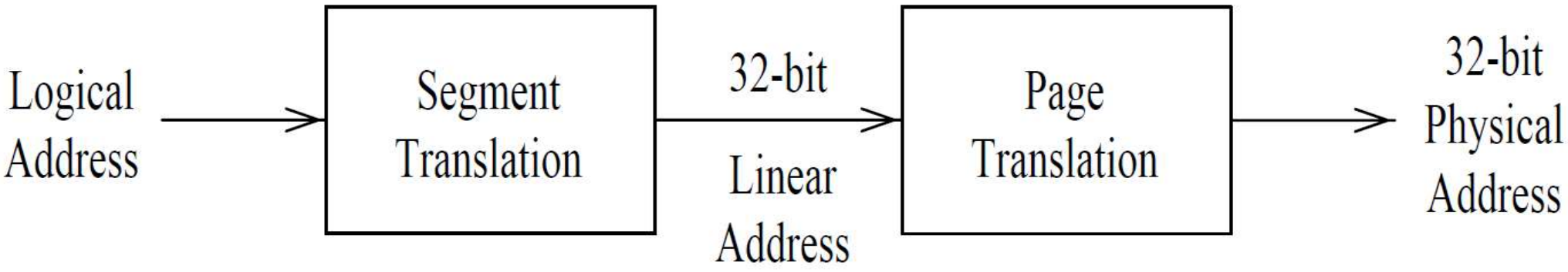
Real Mode Memory Architecture (cont'd)

Six segments of the memory system in real mode



Protected Mode Architecture

- In protected mode, Pentium supports
  - * More sophisticated segmentation
    - » Segmentation can be made invisible (flat model)
  - * Paging for virtual memory
    - » Paging can be turned off

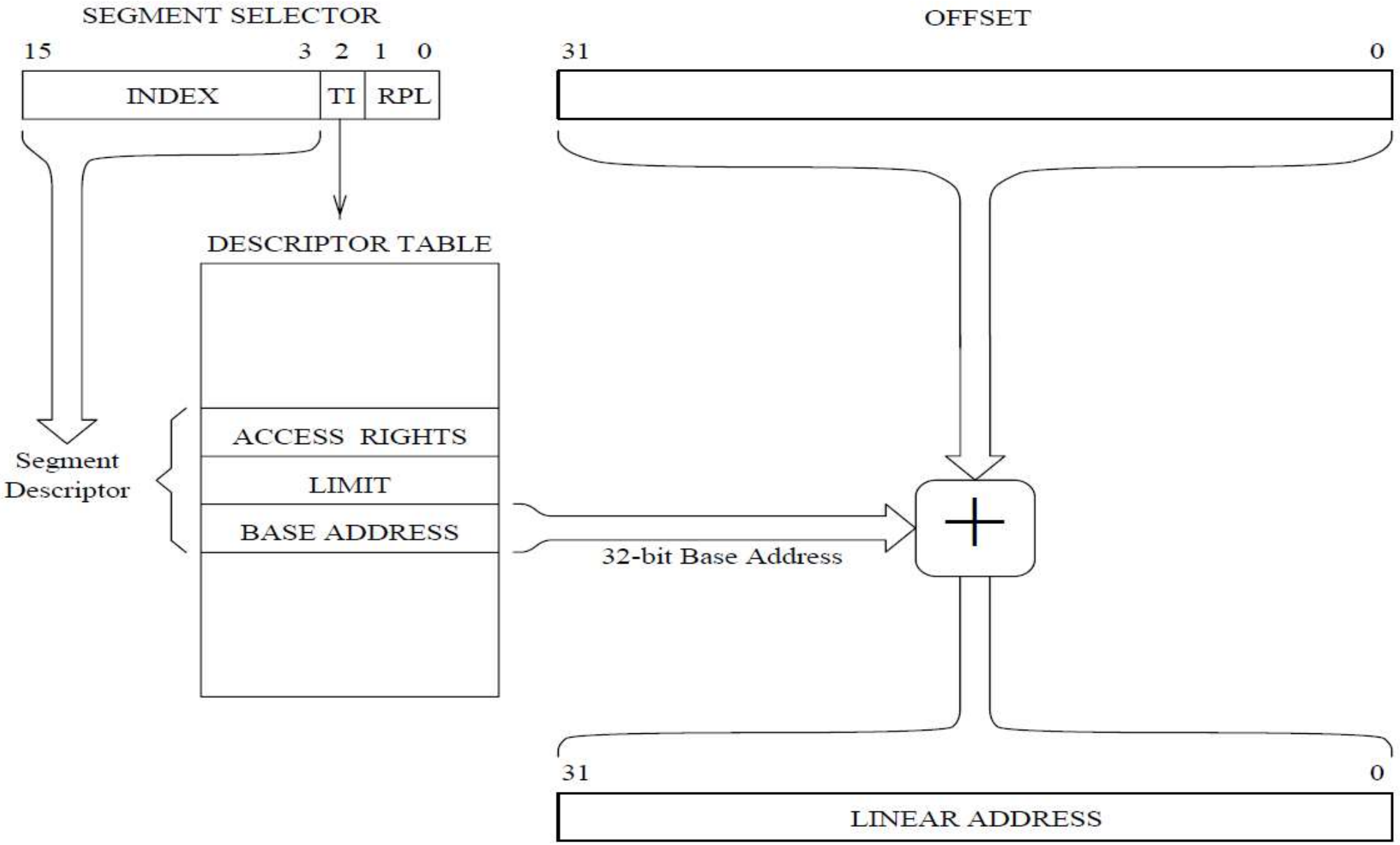


Protected Mode Architecture (cont'd)

- Uses 16-bit segment registers
  - » Real or protected mode, segments registers are 16-bit wide
- Segment register provides an index into a segment descriptor table
  - » Does not provide segment start address as in the real mode

Protected Mode Architecture (cont'd)

Protected mode segment translation



Protected Mode Architecture (cont'd)

- Segment registers have two parts:
  - Visible part
    - This is the 16-bit part we talked about
  - Invisible part
    - Stores the segment descriptor associated with the index
      - ✈Keeps the base address, size, access rights etc.

Visible Part	Invisible Part	
Segment Selector	Segment Base Address, Size, Access Rights, etc.	CS
Segment Selector	Segment Base Address, Size, Access Rights, etc.	SS
Segment Selector	Segment Base Address, Size, Access Rights, etc.	DS
Segment Selector	Segment Base Address, Size, Access Rights, etc.	ES
Segment Selector	Segment Base Address, Size, Access Rights, etc.	FS
Segment Selector	Segment Base Address, Size, Access Rights, etc.	GS

Protected Mode Architecture (cont'd)

- Segment descriptor provides attributes of a segment

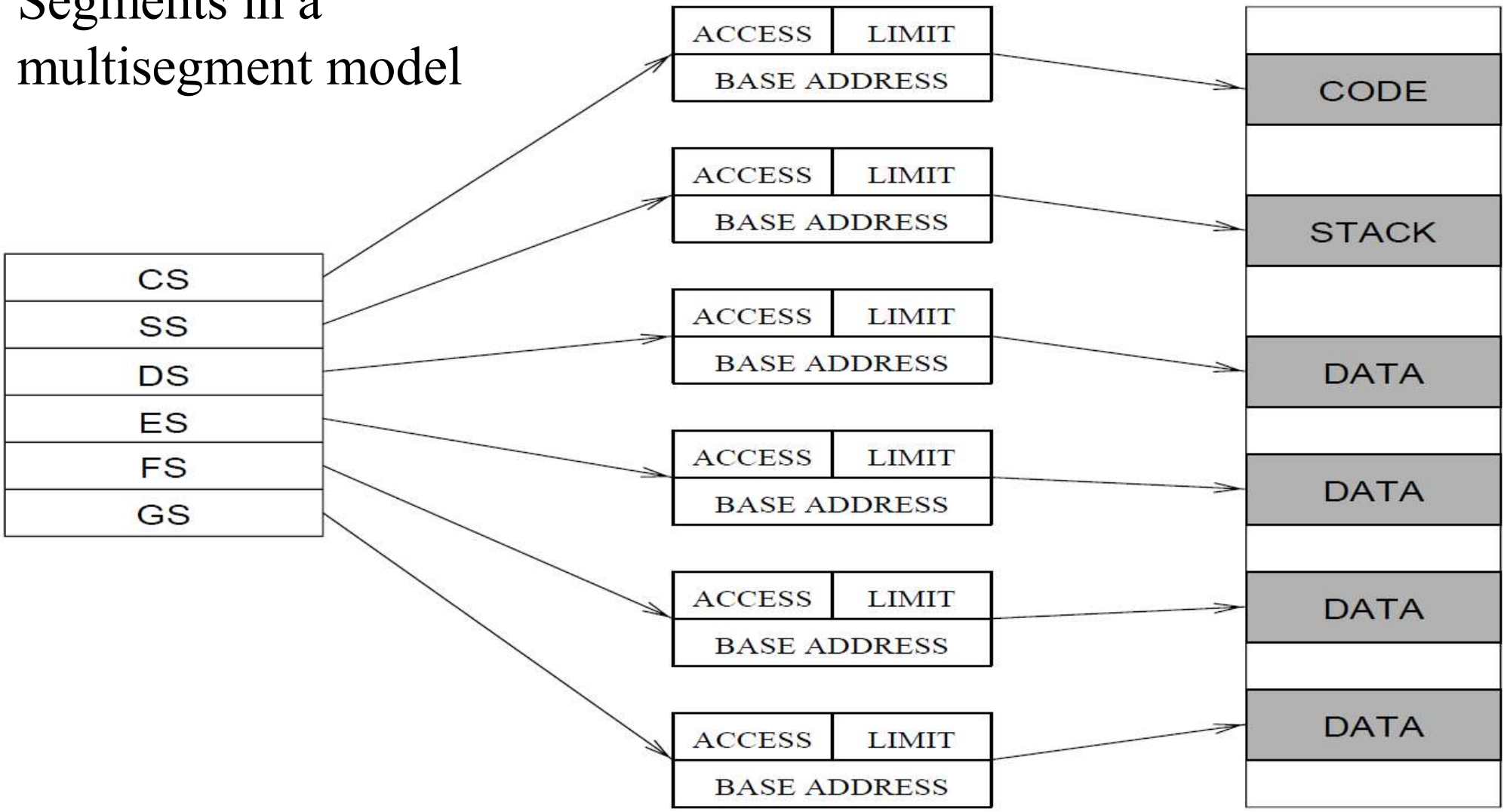
» 32-bit base address

» 20-bit segment size

» Control and status information

Protected Mode Architecture (cont'd)

Segments in a multisegment model



Mixed Mode Operation

- Real mode

» Default: 16-bit model

- Protected mode

» Default: 32-bit model

- Default size is indicated by D bit of the segment descriptor (*D=0,* 16-bit instructions,.

  D=1, 32-bits are assumed.)

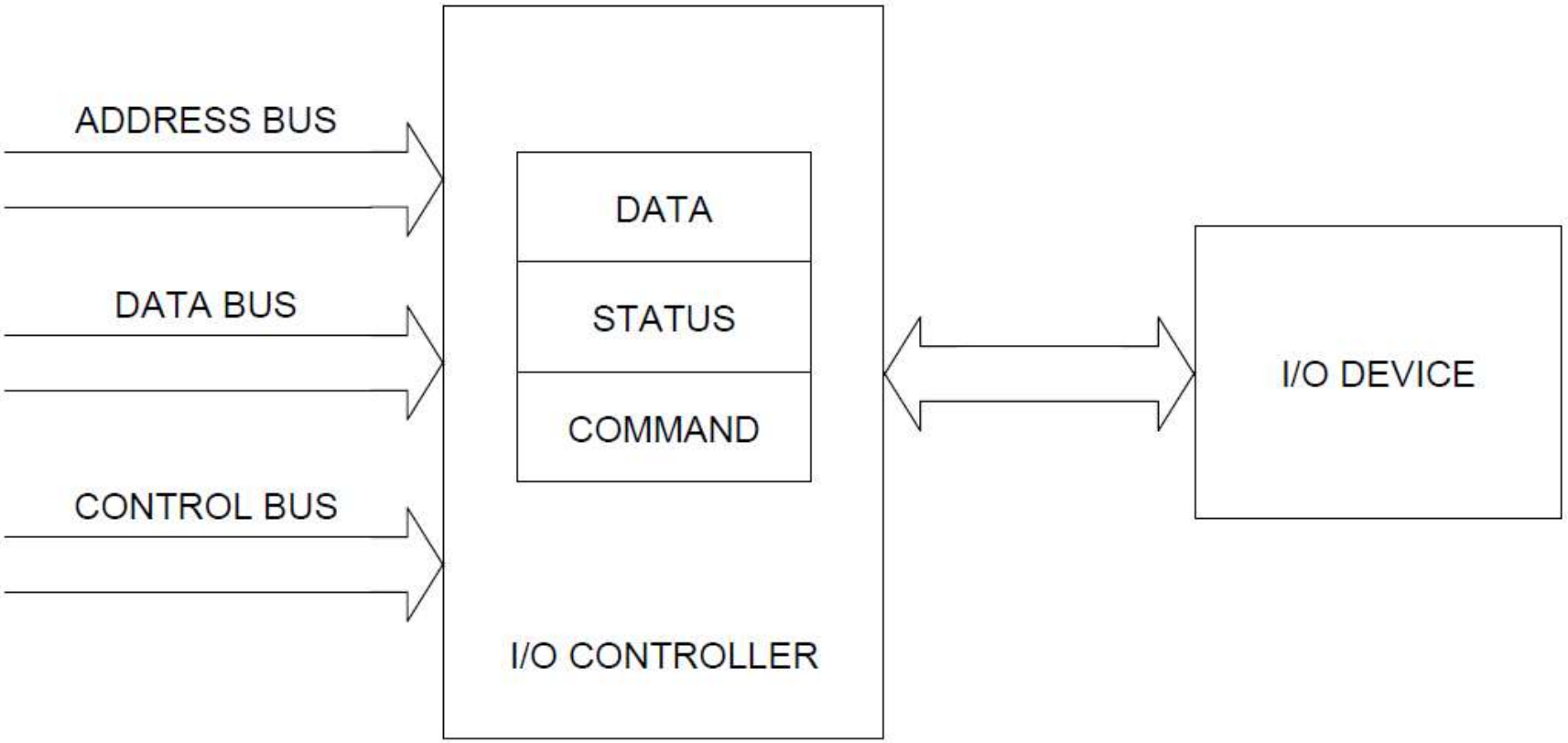
- We can mix 16- and 32-bit data and addresses in either real or protected mode

Input/Output

- I/O controller provides the necessary interface to I/O devices

» Takes care of low-level, device-dependent details

» Provides necessary electrical signal interface



Input/Output (cont'd)

- Processor and I/O interface points for exchanging data are called *I/O ports*
- Two ways of mapping I/O ports
  - Memory-mapped I/O
    - » I/O ports are mapped to the memory address space
    - – Reading/writing I/O is similar to reading/writing memory
      - ✈ Can use memory read/write instructions
    - » Motorola 68000 uses memory-mapped I/O
  - Isolated I/O

» Separate I/O address space

» Requires special I/O instructions (like **`in`** and **`out`** in Pentium)

» Intel 80x86 processors support isolated I/O

Input/Output (cont'd)

- Pentium I/O address space
  - Provides 64 KB I/O address space
  - Can be used for 8-, 16-, and 32-bit I/O ports
  - Combination cannot exceed the total I/O address space
    - can have 64 K 8-bit ports
    - can have 32 K 16-bit ports
    - can have 16 K 32-bit ports
    - A combination of these for a total of 64 KB